Marks

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T. Y. B. Tech (ECE AI-ML)

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Experiment No: 05

Name of the Experiment: Implementation of the Apriori algorithm.

Performed on:

Submitted on:

Teacher's Signature with date

Aim: To implement the **Apriori algorithm** to discover frequent itemsets and generate strong association rules from a **market basket dataset** using Python.

Pre-requisite: Concept of Data Mining, Knowledge of Python programming, Understanding of Support, Confidence, and Lift, Market Basket Analysis.

Objective:

- To understand Association Rule Mining
- To generate frequent itemsets
- To calculate support, confidence, and lift
- To identify strong relationships between items.
- To implement Apriori using Python library

Components and equipment required:

PC with windows 7 or onwards and Anaconda distribution with Python 3.7

Theory:

Apriori algorithm is a standard algorithm in data mining. It is used for mining familiar item sets and relevant association rules. It is developed to perform on a database including a lot of transactions, for example, things brought by shoppers in a store. It is very essential for effective Market Basket Analysis and it helps in understanding which items customers will buy together. It has also been used in the field of healthcare for the discovery of adverse drug reactions. In the Apriori algorithm, frequent k-itemsets are iteratively created for k=1,2,3, and so on… such that k-itemset is created by using prior knowledge of (k-1) itemset. For e.g 3-itemset is generated with prior information of 2-itemset, 4-itemset is created

with prior knowledge of 3-itemset and so on. Because it uses prior knowledge of (k-1) itemset for creating k-itemset, it is named as Apriori algorithm.

Short comings of Apriori Algorithm

The two major shortcomings of the Apriori algorithm are:
The size of candidate itemsets could be extremely large.
High costs on counting support since we have to scan the itemset database over and over again.

Association Rule Mining

Association Rule Mining is a data mining technique used to find relationships or patterns among items in large transactional datasets.
It is widely used in:

- Market basket analysis
- Recommendation systems
- Sales analysis
- Customer behaviour analysis

Apriori Algorithm

Apriori is a classical algorithm used to find:

1. Frequent itemsets
2. Association rules

It works on the principle:
"If an itemset is frequent, all its subsets must also be frequent."

Support

Frequency of occurrence of an itemset.

$$Support(A) = \frac{Transactions\ containing\ A}{Total\ transactions}$$

Confidence

Probability that item B is purchased when A is purchased.

$$Confidence(A \rightarrow B) = \frac{Support(A \cup B)}{Support(A)}$$

Lift

Measures strength of association.

$$Lift = \frac{Confidence}{Support(B)}$$

- Lift > 1 → Positive relation
- Lift = 1 → No relation
- Lift < 1 → Negative relation

Algorithm:

1. Load the market basket dataset.
2. Convert the dataset into a list/transaction format.
3. Apply data cleaning and preprocessing on the data.
4. Flatten the dataset if required using the ravel function.
5. Set minimum thresholds:

- Minimum Support
- Minimum Confidence
- Minimum Lift
- Minimum Length

6. Apply Apriori algorithm to generate frequent itemsets.
7. Generate association rules.
8. Convert generator results into list.
9. Display number of strong rules.
10. Print rules with support, confidence, and lift.
11. Print the top 10 maximum sold items
12. Print the top 10 minimum sold items

Conclusion:

Post Lab Questions:

1. Why is Apriori called an iterative algorithm?
2. What is Association Rule Mining?
3. What happens if minimum support is set very low?
4. Explain Market Basket Analysis with an example.

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Q1. Why is apriori called an iterative algorithm.

A1. Apriori is called an iterative algorithm because it works level by level repeatedly scanning the database & refining results in each step.

Iteration process:

1: Find all frequent 1- itemset (items that ~~support~~ satisfy minimum support)

2: use frequent 1- itemset to generate candidate 2- itemset, then check their support.

3: Use frequent 2- itemset to generate 3-itemsets and so on.

The process repeats until no more frequent itemset can be found.

Each iteration:

- generates layer itemset
- prunes infrequent ones using the apriori property.
- scans the database again.

∴ It is called an iterative algorithm because it gradually builds solution across multiple passes.

Q2. What is Association Rule Mining?

A2. Associate rule mining is a data mining technique used to discover intersecting relationships, patterns or associations between items in large data set.

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It contains:

1. Item set : a collection of items
2. support : how frequently a itemset appears
3. confidence : how often the rule is true
4. lift : strength of association beyond chance.

ALH is widely used in retail, marketing, recommendation systems & web analytics.

Q3. What happens if minimum support is set very low.

A3. A minimum support threshold is very low, the algorithm will generate too many frequent item sets.

Consequences:

1. Huge number of rules
2. High computational cost
3. Noisy + unimportant patterns
4. Difficult interpretation.

∴ overall support lowers the efficiency & usefullness also reduces.

Q4. Explain market Basket Analysis with an example.

A4 Market basket analysis is a practical application of association rule mining used to analyse purchasing behaviour.

Goal:

To find items frequently bought together:

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Transaction	Items
T1	milk, bread
T2	milk, diapers, beetroot
T3	Bread, Butter
T4	Milk, Butter, bread
T5	Bread, butter.

∴ Observed: Bread → Butter

When bread is purchased then butter is also purchased.

∴ Improves sales prediction, user selling.